

University of Michigan Survival Flight Clinical Guideline:E-4

Toxicological Emergencies

Poisoned patients, whether stemming from intentional or accidental etiology, often require prompt and aggressive care. Management of these patients during transport by Survival Flight personnel includes a general approach in addition to specific actions directed at a particular toxin or toxins involved, once identified. The following clinical practice guidelines will assist in formulating differential diagnoses and clinical decision making in order to provide safe transport to tertiary care.

CREW SAFETY IS PARAMOUNT. ENSURE EXTERNAL DECONTAMINATION PRIOR TO LIFT OFF FROM REFERRING INSTITUTION (if applicable)

Oxygen by face mask or nasal cannula (if applicable)

Cardiac monitor (12 Lead EKG)¹

Serum or capillary blood glucose

At least (2) vascular access sites

Results of initial laboratory and radiographic findings²

Concurrent Guidelines:

C-4: Shock

A-1: Airway Management

D-1: Neurological Emergencies

E-2: Hyperkalemia

Useful Phone Numbers

Poison Control: 1(800)222-1222

Chemtrec: 1(800)262-8200 (Haz Mat)

Figure 1: Additional Laboratory Values Useful in Toxic Ingestions and Significance:

Anion Gap (AG)³:

$$[\text{Na}] - ([\text{Cl}] + [\text{HCO}_3])$$

A value > 12 mEq/L is diagnostic for a metabolic derangement

Oxygen Saturation Gap:

exists when there is >5% difference between SaO₂ (ABG) and SpO₂ (pulse oximeter). Toxins associated with this condition may include: CO, methemoglobin, cyanide and hydrogen sulfide

Figure 2: Gastric Decontamination Indications

1. Clinical benefit is most likely if administered within 1 hour of ingestion. Exceptions to this include:

- evidence of continued medication absorption
- ingestion of enteric-coated or sustained-release preparations

2. Consider the need for definitive airway protection as vomiting and aspiration are possible. Likewise, intubate before charcoal administration if deterioration is anticipated based upon ingested toxin.

3. Contraindications to the use of Activated Charcoal as a gastric decontaminant include:

- sedated patients with an unprotected airway
- ingestion of caustic substances or hydrocarbons
- suspected esophageal or gastric perforation
- emergency endoscopy planned at receiving institution

Activated Charcoal contraindications - PHAILS mnemonic:

P: Pesticides
H: Heavy metals, hydrocarbons
A: Acids, alkalis, alcohols
I: Iron
L: Lithium
S: Solvents

Notes:

1. Pay particular attention to prolonged QRS and QT intervals in the setting of tricyclic antidepressant (TCA) and / or cardiac drug toxicity, and peaked or flattened T waves secondary to hyper- or hypokalemia respectively (Refer to E-2: Hyperkalemia, for further guidance). Rightward axis deviation has also been associated with TCA toxicity.

2. Additional laboratory values that may be useful include: anion gap, osmol gap and oxygen saturation gap (Figure 1)

3. Causes of Anion Gap Acidosis:

CAT MUDPILES mnemonic:

C — Carbon monoxide, cyanide
A — Acetaminophen, alcoholic ketoacidosis

T — Toluene

M — Methanol

U — Uremia (chronic kidney failure)

D — Diabetic ketoacidosis

P — Paracetamol, Propylene glycol

I — Infection, Iron, Isoniazid (which can cause lactic acidosis in overdose), Inborn errors of metabolism (an especially important consideration in pediatric patients)

L — Lactic acidosis

E — Ethylene glycol

4. In addition to management of a given shock state, Sodium Bicarbonate therapy should be initiated for specific conditions causing metabolic derangements (e.g., known tricyclic antidepressant (TCA) overdose and salicylate toxicity). Refer to the "Specific Therapy" section for dosing and administration guidance.

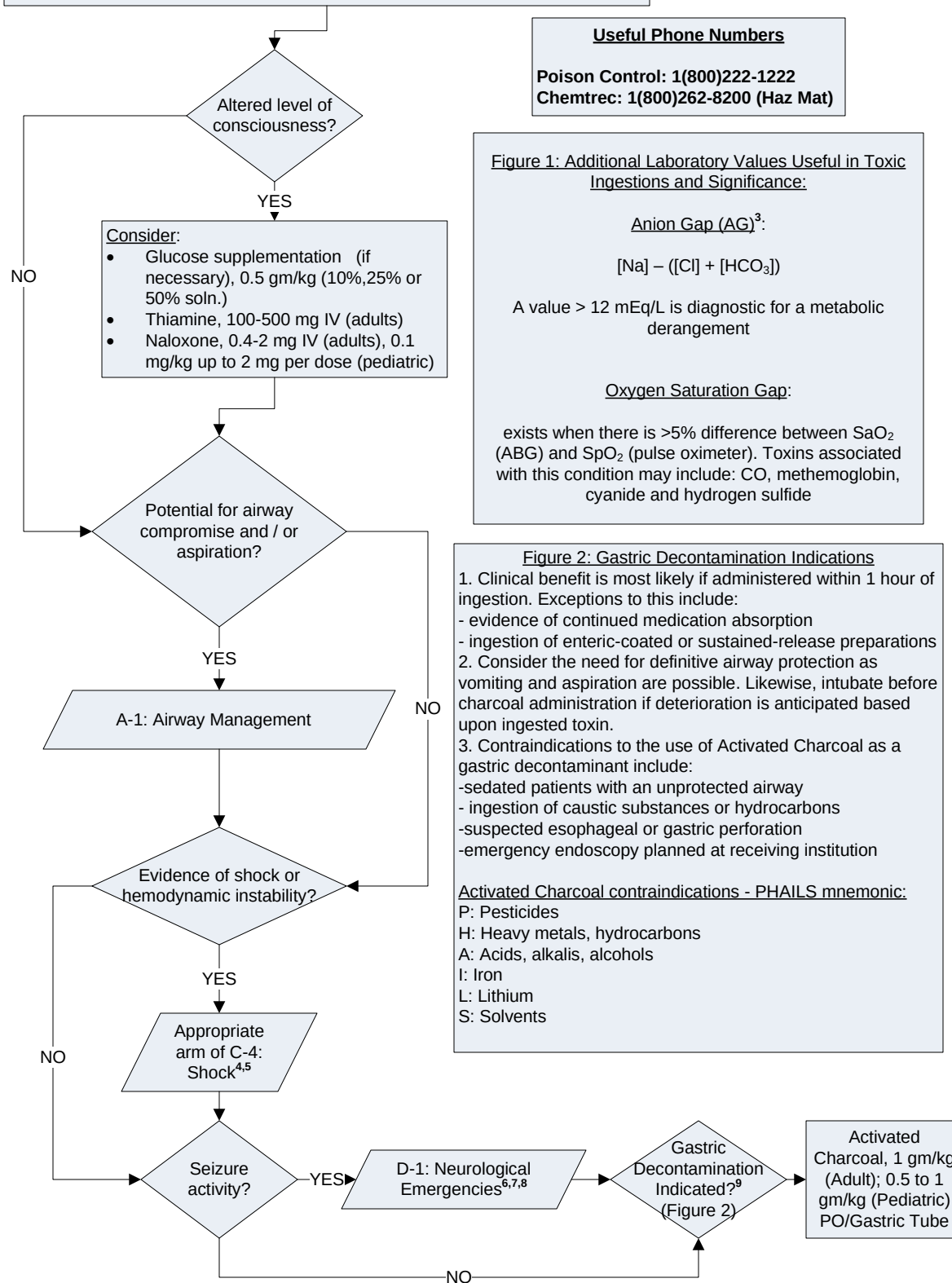
5. Cardiac medications such as beta and calcium channel blocker overdose will require specific therapy in addition to managing the identified shock state. Refer to the "Specific Therapy – Cardiac Drugs" section for dosing and administration guidance.

6. Benzodiazepines considered 1st line treatment followed by Barbiturates

7. If Isoniazid ingestion – give pyridoxine

8. Phenytoin is contraindicated in seizure activity due to ingestion.

9. Consult with medical control and poison control for further guidance.



COMMON TOXIDROMES

Toxidrome	Common Findings	Other Signs / Symptoms	Possible Agents	General Interventions
Opioid	CNS and Respiratory Depression; Constricted Pupils	Hypothermia, Bradycardia	Heroin, Morphine, Fentanyl, oral opiates, clonidine (acts like an opiate in overdose)	Naloxone; Ventilatory Support
Sympathomimetic	Agitation, dilated pupils, diaphoresis, tachycardia, HTN, hyperthermia (Diaphoresis distinguishes from Anticholinergic)	Seizures, rhabdomyolysis, ACS	Cocaine, Amphetamines, MDMA	IV hydration, large quantities of benzodiazepines, external cooling
Cholinergic	Salivation, Lacrimation, Urination, Diaphoresis, Diarrhea, Vomiting, Emesis(SLUDGE) Constricted pupils, muscle fasciculations weakness	Bradycardia, respiratory failure, seizures, paralysis	Insecticides, some nerve agents	Decontamination, airway protection, ventilation, large doses of atropine, pralidoxime (2-PAM)
Anticholinergic	Altered LOC, hyperthermia, dry mucous membranes, dilated pupils, dry / flushed skin, urinary retention, decreased bowel sounds	Seizures, dysrhythmias, rhabdomyolysis	Scopolamine, atropine, tricyclic antidepressants (TCA), Diphenhydramine	Supportive management, sedation with benzodiazepines, external cooling
Salicylates	Altered mental status, respiratory alkalosis, metabolic acidosis, tinnitus, hyperpnea, tachycardia, diaphoresis, nausea, vomiting	Ketonuria, cerebral edema, acute lung injury	Aspirin, Oil of Wintergreen	Activated Charcoal, Alkalinization of the Urine, Potassium Repletion. AVOID INTUBATION IF POSSIBLE. If intubation necessary, match patient's inherent minute ventilation
Serotonin (Autonomic Hyperactivity)	Agitation, anxiety, disorientation, restlessness, excitement, tachycardia, sweating, dilated pupils, hyperreflexia, inducible and ocular clonus, HTN, <u>hyperthermia, peripheral hypertonicity, truncal rigidity (indicates life threat with high risk for respiratory failure)</u>	Headache, shivering, metabolic acidosis, rhabdomyolysis, seizures, DIC, coma	SSRI / SNRI's, MAOI's, TCA's, CND Stimulants, Dextromethorphan, Metoclopramide, Ondansetron	Activated Charcoal, Supportive care. Shock treated with pressors (epinephrine, norepinephrine), benzodiazepines for agitation/hyperactivity

Common Antidotes

<u>Ingested Agent</u>	<u>Antidote</u>	<u>Adult Dose</u>	<u>Pediatric Dose</u>
Acetaminophen	N-Acetylcystine (NAC)	Adult/Ped: 150 mg/kg IV over 15-60 minutes. Then 12.5 mg/kg/hr for 4 hours (50 mg/kg in 500 ml run over 4 hours). Then 6.25 mg/kg/hr for 16 hours (100 mg/kg in 1000 ml run over 16 hours). (NAC infusion may be continued at 6.25mg/kg/hr if transaminase levels remain elevated) For peds, adjust infusate volume.	
Ca²⁺ Channel Blockers β – Blockers	Calcium Chloride (10%)	Adult/Ped: 20 mg/kg IV bolus (MAX 2 Grams), then 20-40 mg/kg/hr gtt	
	High Dose Insulin Therapy	Adult/Ped: 1 unit/kg IV bolus, then 1 unit/kg/hr	
	Lipid Emulsion Therapy	Per box below	
Digoxin	Digoxin Fab (DigiFab)	<u>Arrest/Near Arrest:</u> 10 vials slow IVP <u>All others:</u> Consult poison control (1-800-222-1222) & administer over 30 mins	5 vials slow IVP
Ethylene Glycol, Methanol	Ethanol (10%)	Adult/Ped: 10 ml/kg (10% soln) over 30 min, then 1.2 ml/kg/hr IV	
	Fomepizole	Adult/Ped: 15 mg/kg IV, then 10 mg/kg every 12 hours	
Opioids, Clonidine	Naloxone	0.4 to 2 mg IV (titrate to effect)	0.05 to 0.1 mg/kg IV (titrate to effect)
Cholinergic agents (pesticides, some nerve agents)	Atropine	1-5 mg IV, double dose q 5 min Once stable, infusion at 10-20% of cumulative dose (e.g. If needed 18 mg Atropine, start infusion at 1.8-3.6 mg/hr)	0.05-0.1 mg/kg IV, double dose q 5 min
	Pralidoxime (2-PAM)	Adult/Ped: 30 mg/kg bolus over 20 minutes (max 2000mg) then 8-10 mg/kg/hr infusion (max 500 mg/hr)	
Isoniazid (INH)	Pyridoxine	5 grams IV	70 mg/kg IV (max 5 grams)
Salicylates; Tricyclic Antidepressants (TCA)	Sodium Bicarbonate Lipid Emulsion Therapy	Adult/Ped: 1-2 mEq/kg IV; then 2 mEq/kg/hr infusion Per box below	
Oral Hypoglycemics (sulfonylurea-induced hypoglycemia)	Octreotide	100 mcg SQ (1-3 doses) OR 50-125 mcg/hr (cont. infusion)	2 mcg/kg SQ

**Antihypertensives (CCB,BB, ACEi),
Tricyclic Antidepressants, or
Polysubstance Overdose
UNRESPONSIVE TO OTHER TX**

Angiotensin II
(Giapreza)

MUST CONSULT MEDICAL CONTROL
(Dosing per C4.2 Distributive Shock)

20% Lipid Emulsion Therapy

Indications

Regional Anesthetics (e.g., bupivacaine)

Rescue therapy:

- Ca²⁺ Channel Blockers
- β – Blockers
- Lipophilic Cardiotoxins (e.g. quetiapine, amitriptyline, bupropion)

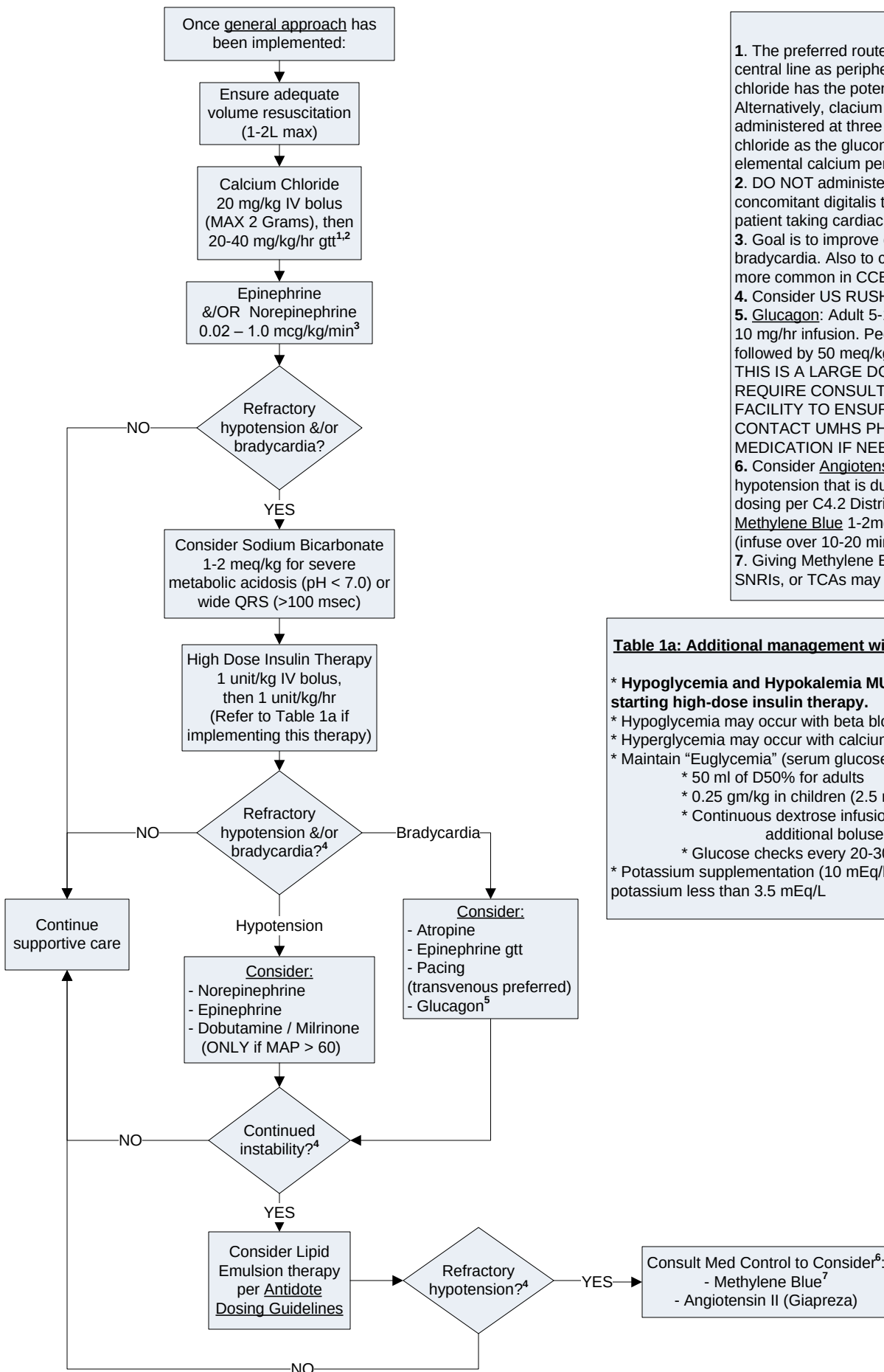
Dosing

1.5 ml/kg IV bolus followed by infusion at 0.25 ml/kg/min x 60 min
(Consider consult w/ Poison Control as this dosing may be adjusted based on drug(s) ingested, amount, & pt condition)

Antithrombotic Agent Reversal

<u>Ingested Agent</u>	<u>Antidote</u>	<u>Adult Dose</u>	<u>Administration</u>
<u>Vitamin K Antagonist</u> Warfarin (Coumadin)	Vitamin K Prothrombin complex concentrate (PCC) Kcentra	10 mg IVPB (MUST be given with PCC or FFP) <u>Dosing based on INR level:</u> INR 1.4-1.9: 15 units/kg (MAX 1,500 units) 2.0-3.9: 25 units/kg (MAX 2,500 units) 4.0-6.0 35 units/kg (MAX 3,500 units) > 6.0 50 units/kg (MAX 5,000 units)	Infuse over 30 min Kcentra is supplied in 500 unit/20ml vials. Doses rounded to nearest whole vial. 1. Obtain & empty 250 ml NS bag 2. Inject reconstituted Kcentra into EMPTY 250 ml bag 3. Administer via IVPB at a rate of 7.2 ml/kg/hr (~3 units/kg/min) to a MAX rate of 500 ml/hr
If PCC unavailable: Fresh Frozen Plasma (FFP)		10-20 ml/kg	
<u>Direct Thrombin Inhibitors</u> Dabigatran (Pradaxa)	Activated Charcoal (Only if ingested < 2 hours ago) Idarucizumab (Praxbind)	50 Grams 5 Grams rapid IV bolus	Must use dedicated IV line
	If Praxbind unavailable: Kcentra	50 units/kg (MAX 5,000 units)	Administration per note above
<u>Factor Xa Inhibitors</u> Rivaroxaban (Xarelto) Apixaban (Eliquis)	Activated Charcoal (Only if ingested in last 8 hours)	50 Grams	
* Intracranial Hemorrhage:	Andexanet (Andexxa) <u>(-OR- Kcentra per below)</u>	Dosing based on time since last dose of Xarelto/Eliquis Last dose greater than 8 hours ago: - IV bolus: 400 mg / 250 ml NS infused over 15 mins - Then drip: 480 mg / 250 ml NS infused over 2 hrs (125ml/hr) Last dose less than 8 hours ago or UNKNOWN - IV bolus: 800 mg / 250 ml NS infused over 30 mins - Then drip: 960 mg / 250 ml NS infused over 2 hrs (125 ml/hr)	* Must use 0.2 micro in-line filter *
* Life threatening bleeding of another site (NOT ICH):	Kcentra	50 units / kg (MAX 5,000 units)	Administration per note above
	If bleeding unresponsive to Kcentra, consider Feiba NF or Factor VIIa (Novoseven) PER MEDICAL CONTROL		
<u>Thrombolytics</u> Alteplase (Activase, tPA) Cenectopase (TNKase) Reteplase (Retavase) Urokinase (Abbokinase) Streptokinase (Streptase)	Cryoprecipitate Tranexamic acid (TXA) (-OR- Aminocaproic Acid [Amicar]) If bleeding persists consider FFP & Platelet transfusions	10 units 1 Gram / 100 ml NS infused over 10 mins (5 Grams / 250 ml NS infused over 20-60 min)	
<u>Unfractionated Heparin</u> * Heparin DRIP	Protamine Sulfate	1 mg IV for every 100 units of Heparin received over last 3 hours (MAX 50 mg) (example: heparin gtt at 1,000 units/hr would require 30mg Protamine)	Infuse over 10 mins (Rapid administration will cause hypotension & bradycardia)
* Heparin SubQ	Protamine Sulfate	1 mg IV for ever 100 units of Heparin received over last 8 hours (MAX 50 mg)	Infuse over 10 mins
<u>Low Molecular Weight Heparins</u> Enoxaparin (Lovenox)	Protamine Sulfate	Last dose less than 8 hours ago: - 1mg Protamine (MAX 50 mg) for every 1 mg Enoxaparin Last dose greater than 8 hours ago: ½ of above dose	Infuse over 10 mins (Rapid administration will cause hypotension & bradycardia)
Daltepain (Fragmin) Tinzaprin (Innohep)	Protamine Sulfate	1 mg Protamine (MAX 50 mg) for every 100 units Daltepain/Tinzaprin	Infuse over 10 mins
<u>Anti-Platelet Agents</u> ASA Clopidogrel (Plavix) Ticagrelor (Brilinta) Prasugrel (Effient) Dipyridamole (Aggrenox) Ticlopidine (Ticlid) Eptifibatide (Integrilin)	Desmopressin (DDAVP) Platelet transfusion (may be helpful especially with ASA / Plavix / Effient) If bleeding persists, consider Tranexamic acid (TXA) 1 Gram / 100 ml NS infused over 10 mins	0.4 mcg/kg (MAX 40 mcg) in 100 ml NS 2 units	Infuse over 15 mins

Specific Considerations #1: Calcium Channel Blocker and Beta Blocker Toxicity



NOTES:

1. The preferred route of administration is via a central line as peripheral extravasation of calcium chloride has the potential to cause tissue necrosis. Alternatively, calcium gluconate may be administered at three (3) times the dose of calcium chloride as the gluconate preparation has 1/3 of the elemental calcium per ml.
2. DO NOT administer calcium in the presence of concomitant digitalis toxicity, OR hyperkalemia in a patient taking cardiac glycoside medication.
3. Goal is to improve cardiac contractility & bradycardia. Also to correct vasodilation which is more common in CCB OD.
4. Consider US RUSH exam to guide treatment
5. Glucagon: Adult 5-10 mg IV bolus followed by 1-10 mg/hr infusion. Ped: 50-150 mcg/kg IV bolus followed by 50 mcg/kg/hr infusion. NOTE THAT THIS IS A LARGE DOSING REGIMEN AND MAY REQUIRE CONSULTATION WITH REFERRING FACILITY TO ENSURE ADEQUATE SUPPLY. CONTACT UMHS PHARMACY TO OBTAIN MORE MEDICATION IF NEEDED.
6. Consider Angiotensin II (Giapreza) for refractory hypotension that is due in part to vasoplegic shock - dosing per C4.2 Distributive Shock. – OR -- Methylene Blue 1-2mg/kg in 100 ml 0.9 NS IV (infuse over 10-20 min)
7. Giving Methylene Blue to pts taking SSRIs, SNRIs, or TCAs may cause Serotonin Syndrome

Table 1a: Additional management with high-dose insulin therapy

- * Hypoglycemia and Hypokalemia **MUST** be corrected prior to starting high-dose insulin therapy.
- * Hypoglycemia may occur with beta blocker toxicity
- * Hyperglycemia may occur with calcium channel blocker toxicity
- * Maintain "Euglycemia" (serum glucose > 150 mg/dl):
 - * 50 ml of D50% for adults
 - * 0.25 gm/kg in children (2.5 ml/kg 10% Dextrose solution)
- * Continuous dextrose infusion of 0.5 to 1 gm/kg/hr with additional boluses as needed.
- * Glucose checks every 20-30 minutes during therapy
- * Potassium supplementation (10 mEq/hr peripherally) for a serum potassium less than 3.5 mEq/L

Specific Considerations #2: Ketamine, Cocaine and Amphetamine Abuse

Once general approach has been implemented:

Ensure large-bore IV access x 2 and foley catheterization for urine output monitoring and alleviation of retention.

Consider Benzodiazepine administration per S-1: Patient Comfort and Safety for agitation and hypertension¹

Refractory Hypertension after adequate benzo administration?²

YES

Consider Phentolamine 5 mg IV q 5-15 min PRN

Nitroprusside infusion: initial dose 0.3 mcg/kg/min

Hyperthermic?

YES

Aggressive cooling measures per E-3.1: Heat Related Illness

Seizure activity unresponsive to benzodiazepines?

YES

D-1: Neurological Emergencies³

Evidence of Rhabdomyolysis?^{4,5,6,7}

NO

Volume resuscitation with Lactated Ringers

Evidence of coronary ischemia?

C-3: Acute Coronary Syndromes^{2,8}

Supportive Care

Concurrent Guidelines

S-1: Patient Comfort and Safety
D-1: Neurological Emergencies
E-3: Environmental Emergencies
C-3: Acute Coronary Syndromes

Arrhythmia Management Considerations:

Sinus Tachycardia: sedation, cooling, fluid resuscitation

SVT / AF w/ RVR / A-flutter: Calcium channel blocker

Wide Complex Tachycardia:

- Sodium bicarb: 1-2 meq/kg IV bolus followed by intermittent boluses or infusion (guided by clinical response and serum pH - DO NOT alkalinize above pH 7.55)
- Lidocaine can be considered.
- Torsades: Consider Lidocaine, Magnesium Sulfate, & overdrive pacing

In severe intoxication with persistent CV instability and/or refractory WCT, intralipid can be considered.

- IV 20% lipid 100 ml (1.5ml/kg) bolus over 2-3 minutes followed by infusion of 18 ml (0.25ml/kg) per minute until clinical improvement or max total dose 10ml/kg given.

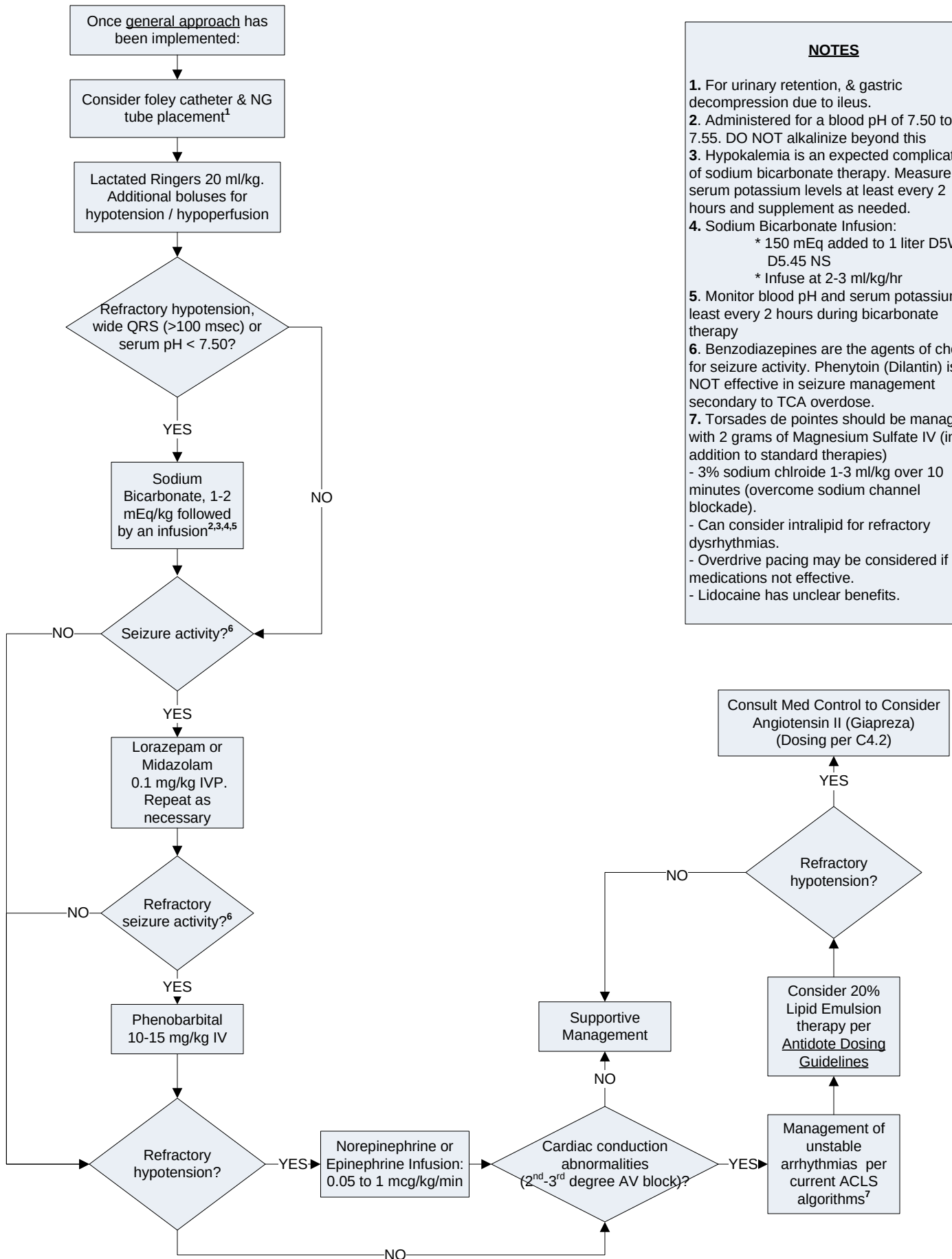
NOTES

1. There is no upper limit for benzodiazepine administration for this type of toxicity. Consider double escalating doses until adequate sedation achieved (e.g. Ativan 2mg, then 4mg, then 8mg...). Airway protection may be necessary.
2. Beta blocking agents contraindicated due to potential for unopposed alpha receptor stimulation, exacerbating hypertension.
3. Antipsychotics (eg., haloperidol, droperidol and chlorpromazine) should be avoided as these agents may lower the patient's seizure threshold.
4. Elevated CPK level (typically 5 x normal or more), increased urine myoglobin, "tea colored" urine, and / or hyperkalemia. If no testing available, suspect rhabdomyolysis with significant agitation and treat early.
5. Target euvoolemia for initial resuscitation of suspected rhabdomyolysis. After initial volume, fluid rate should be reduced to 2.5 ml/kg/hr targeting a minimum urine output of 2 ml/kg/hr.
6. May consider isotonic bicarb infusion. Indicated for: uremic acidosis especially with hyperkalemia, Non-anion gap metabolic acidosis (hyperchloremic metabolic acidosis). Mix 150 meq of sodium bicarb in 1 liter of D5W and run at 150 ml/hr.
7. Avoid succinylcholine as effects can be unpredictable.
8. Caveats in ACS and cocaine intoxication: may consider use of calcium-channel blockers. Labetalol may be safe given both alpha and beta properties. If on IV nitroglycerin, beta blockers may be considered.

Specific Considerations #3: Overdose of Tricyclic Antidepressants (TCA)

NOTES

1. For urinary retention, & gastric decompression due to ileus.
2. Administered for a blood pH of 7.50 to 7.55. DO NOT alkalinize beyond this
3. Hypokalemia is an expected complication of sodium bicarbonate therapy. Measure serum potassium levels at least every 2 hours and supplement as needed.
4. Sodium Bicarbonate Infusion:
 - * 150 mEq added to 1 liter D5W or D5.45 NS
 - * Infuse at 2-3 ml/kg/hr
5. Monitor blood pH and serum potassium at least every 2 hours during bicarbonate therapy
6. Benzodiazepines are the agents of choice for seizure activity. Phenytoin (Dilantin) is NOT effective in seizure management secondary to TCA overdose.
7. Torsades de pointes should be managed with 2 grams of Magnesium Sulfate IV (in addition to standard therapies)
 - 3% sodium chloride 1-3 ml/kg over 10 minutes (overcome sodium channel blockade).
 - Can consider intralipid for refractory dysrhythmias.
 - Overdrive pacing may be considered if medications not effective.
 - Lidocaine has unclear benefits.



Specific Considerations #4: Suspected Serotonin Syndrome

HUNTER CRITERIA for DIAGNOSIS of SEROTONIN SYNDROME

(Ingestion of Serotonin Agent PLUS ONE of the Following):

- Spontaneous clonus
- Inducible Clonus plus agitation, or diaphoresis
- Ocular Clonus plus agitation or diaphoresis
- Tremor plus hyperreflexia
- Hypertonia plus temperature > 38C, plus ocular clonus OR Inducible Clonus

Figure 2: Severity of Toxicity

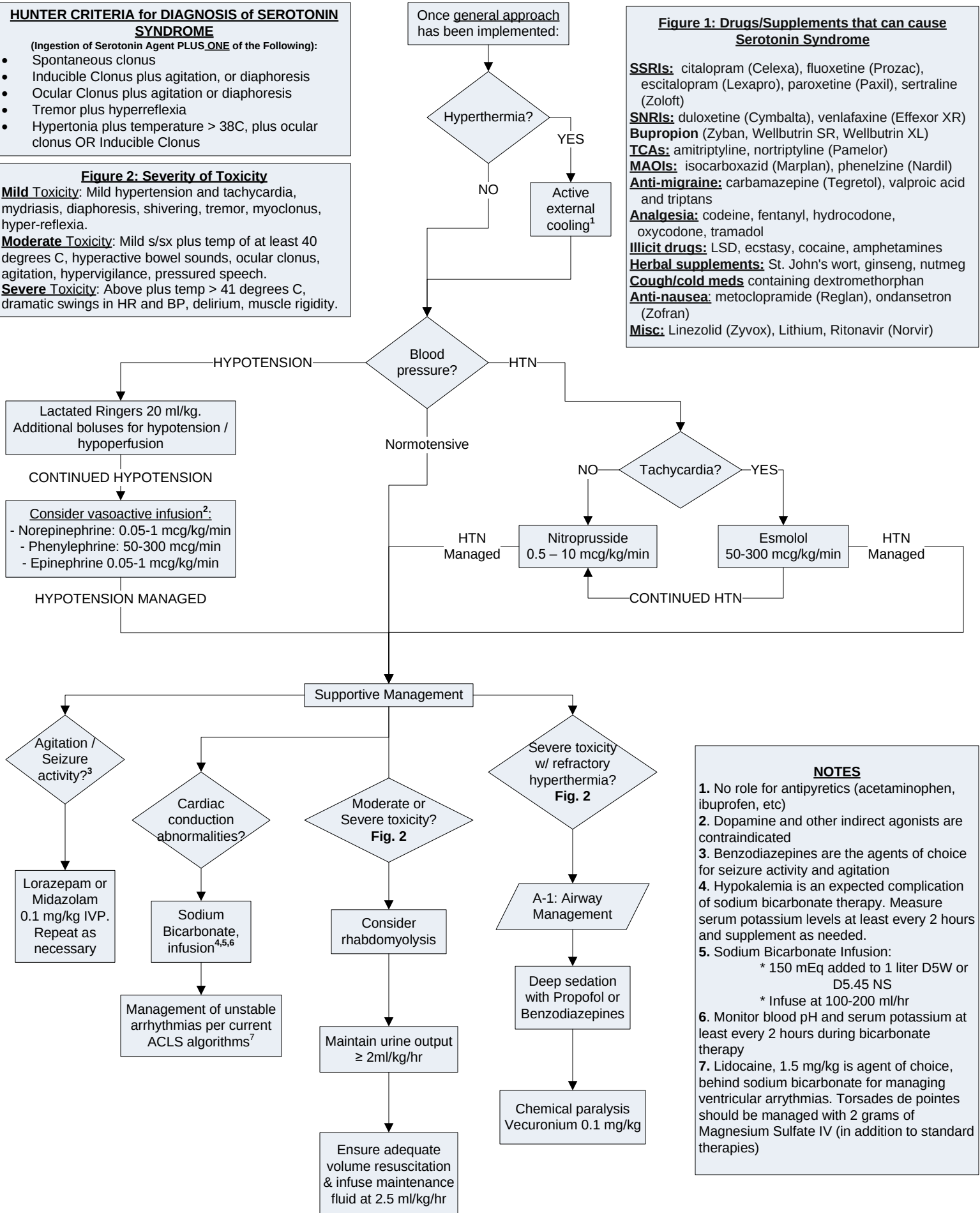
Mild Toxicity: Mild hypertension and tachycardia, mydriasis, diaphoresis, shivering, tremor, myoclonus, hyper-reflexia.

Moderate Toxicity: Mild s/sx plus temp of at least 40 degrees C, hyperactive bowel sounds, ocular clonus, agitation, hypervigilance, pressured speech.

Severe Toxicity: Above plus temp > 41 degrees C, dramatic swings in HR and BP, delirium, muscle rigidity.

Figure 1: Drugs/Supplements that can cause Serotonin Syndrome

SSRIs: citalopram (Celexa), fluoxetine (Prozac), escitalopram (Lexapro), paroxetine (Paxil), sertraline (Zoloft)
SNRIs: duloxetine (Cymbalta), venlafaxine (Effexor XR)
Bupropion (Zyban, Wellbutrin SR, Wellbutrin XL)
TCAs: amitriptyline, nortriptyline (Pamelor)
MAOIs: isocarboxazid (Marplan), phenelzine (Nardil)
Anti-migraine: carbamazepine (Tegretol), valproic acid and triptans
Analgesia: codeine, fentanyl, hydrocodone, oxycodone, tramadol
Illicit drugs: LSD, ecstasy, cocaine, amphetamines
Herbal supplements: St. John's wort, ginseng, nutmeg
Cough/cold meds containing dextromethorphan
Anti-nausea: metoclopramide (Reglan), ondansetron (Zofran)
Misc: Linezolid (Zyvox), Lithium, Ritonavir (Norvir)



NOTES

1. No role for antipyretics (acetaminophen, ibuprofen, etc)
2. Dopamine and other indirect agonists are contraindicated
3. Benzodiazepines are the agents of choice for seizure activity and agitation
4. Hypokalemia is an expected complication of sodium bicarbonate therapy. Measure serum potassium levels at least every 2 hours and supplement as needed.
5. Sodium Bicarbonate Infusion:
 - * 150 mEq added to 1 liter D5W or D5.45 NS
 - * Infuse at 100-200 ml/hr
6. Monitor blood pH and serum potassium at least every 2 hours during bicarbonate therapy
7. Lidocaine, 1.5 mg/kg is agent of choice, behind sodium bicarbonate for managing ventricular arrhythmias. Torsades de pointes should be managed with 2 grams of Magnesium Sulfate IV (in addition to standard therapies)

Specific Considerations #5: Exposure to Pesticides and Nerve Agents with Cholinergic Properties

Common Agents

*Organophosphate pesticides
*Nerve Agents:
* Sarin gas
* VX gas

Common Toxidromes

"SLUDGE"

Salivation
Lacrimation
Urinary Incontinence
Defecation
GI Cramping
Emesis

"DUMBELS"

Diarrhea / Diaphoresis
Urination
Miosis / Muscle Weakness
Bradycardia/Bronchorrhea/Bronchospasm
Emesis
Lacrimation
Salivation

Killer B's: Bradycardia, bronchorrhea, bronchospasm

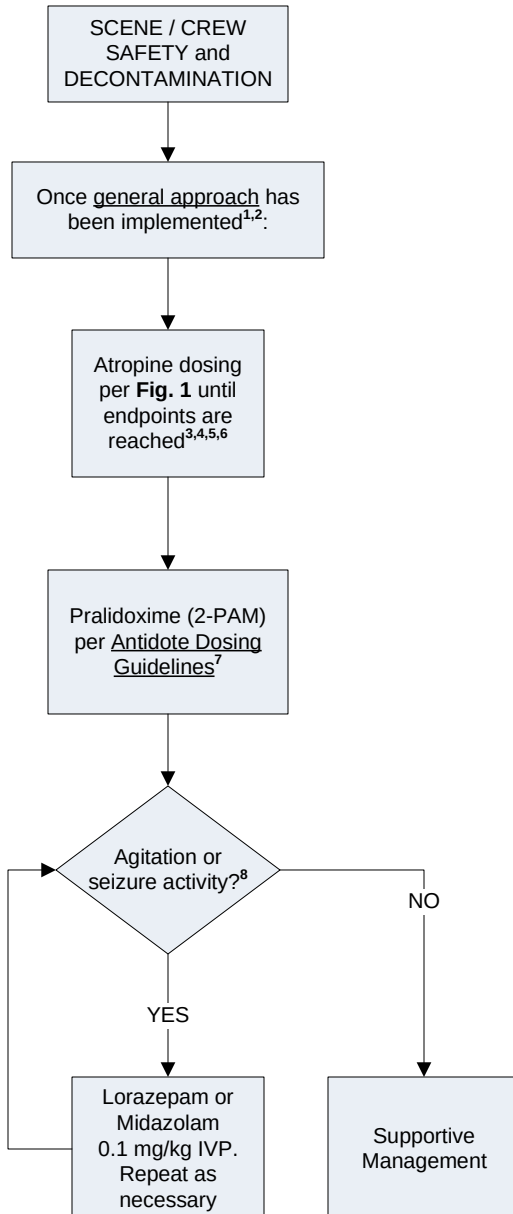


Figure 1: Atropine Dosing

Adult:

Mild-moderate: 1-2 mg bolus, repeat every 3-5 min with double escalating doses

Severe: 3-5 mg bolus, repeat every 3-5 min with double escalating dose

Peds: 0.05-0.1 mg/kg bolus, repeat every 5-10 min with double escalating dose

Infusion: Once stable, infusion at 10-20% of cumulative dose
(If needed 18 mg Atropine, start infusion at 1.8-3.6 mg/hr)

NOTES

1. Early airway management should be considered. Complications will include excessive secretions and higher than normal airway pressures needed for effective ventilation
2. While not absolutely contraindicated, succinylcholine will have a prolonged effect. Consider utilizing rocuronium (zemuron) at the higher dose range.
3. The endpoint to atropine administration is drying of respiratory secretions, improved ventilation **AND** a mean arterial pressure (MAP) > 60 mmHg
4. DOSE EARLY
5. Tachycardia **not** a contraindication for treatment with atropine. Can consider nebulized atropine or ipratropium for respiratory symptom management
6. Beta blockers **absolutely** contraindicated- can worsen intoxication
7. Pralidoxime contraindicated in asymptomatic or minimally symptomatic
8. For seizures, give additional atropine first (may abort seizure), followed by benzos

Specific Considerations #6: Salicylate and Acetaminophen Toxicity

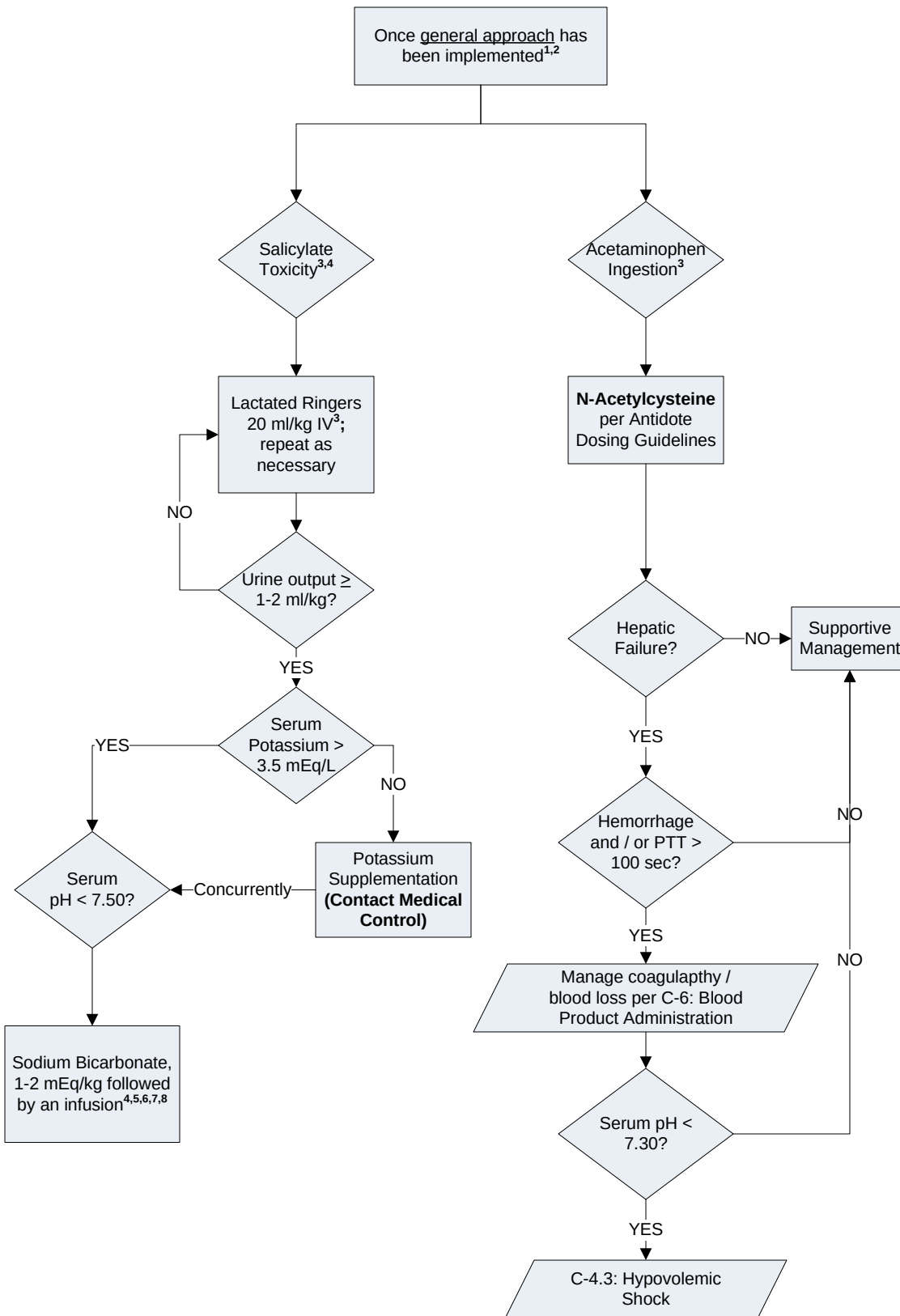
Once general approach has been implemented^{1,2}

Concurrent Protocols:

C-4.3: Hypovolemic Shock
C-6: Blood Product Administration

NOTES:

1. Ensure that a recent blood gas and serum potassium level have been obtained.
2. If intubation and assist ventilation become necessary, it is important to maintain the patient's inherent minute ventilation (Ve). Most patients with severe salicylate intoxication have respiratory alkalosis. Controlling the ventilatory rate to "normal" values decreases serum pH and causes a rapid shift of salicylate in to the central nervous system. Consider giving sodium bicarbonate 1-2 meq/kg prior to induction as a bicarb buffer of acidosis during apneic time.
3. Consider single dose activated charcoal if ingestion < 4 hours or unknown
4. Although rare, if serious bleeding present in the setting of salicylate overdose, FFP is agent of choice. No roll for Vitamin K.
5. Ensure that there are at Least 2 vascular access sites: one for volume resuscitation and one for subsequent infusion of sodium bicarbonate. Both hydration and urine alkalization can be instituted simultaneously.
6. Administered for a blood pH of 7.50 to 7.55. DO NOT alkalize beyond this
7. Hypokalemia is an expected complication of sodium bicarbonate therapy. Measure serum potassium levels at least every 2 hours and supplement as needed.
8. Sodium Bicarbonate Infusion:
* 150 mEq added to 1 liter D5W or D5.45 NS
* Infuse at 2-3 ml/kg/hr
9. Monitor blood pH and serum potassium at least every 2 hours during bicarbonate therapy
10. Except for the fluids used for initial resuscitation, all IV fluids administered should contain a minimum of 50 grams/L of glucose (5% dextrose in water) because hypoglycemia has been implicated in the pathophysiology of salicylate neurologic injury.



Specific Considerations #7: Methemoglobinemia

Methemoglobinemia: altered state of hemoglobin where ferrous ions (Fe^{2+}) of heme are oxidized to ferric state (Fe^{3+}) rendering it unable to transport oxygen

Figure 1: Causes of acquired methemoglobinemia

- **Topical anesthetics:** Benzocaine, Lidocaine, Prilocaine
- **Nitrates:** Inhaled Nitric Oxide, Nitroglycerine, Nitroprusside
- **Antibiotics:** Dapsone, Rifampin, Sulfonamides, Antimalarials (chloroquine)
- **Miscellaneous Meds:** Rasburicase, Metoclopramide
- **Environmental/Occupational:** Well water (nitrate contamination), fertilizers, weed killers, dyes, paints, rubber

Once general approach has been implemented

Supplemental O2
15 lpm NRB

Significant symptoms &/OR
MethHb level > 30%?¹
Fig. 2 & 3

NO

YES

Confirmed or suspected G6PD
deficiency?^{2,3}

NO

YES

Ascorbic acid
(Vitamin C)
2-10 Grams IV
over 15-30 mins

Contact med control for
other possible treatments:
- PRBC transfusion
- Methylene Blue
- Hyperbaric oxygen
therapy

Pt on
Serotonergic
medication?⁴
Fig. 4

YES

NO

Methylene Blue
1-2 mg/kg in 100ml NS
IV over 15-20 mins⁵

Clinical
improvement?

YES

NO

Additional dose of
Methylene Blue
may be given 30-60 min
after first dose^{5,6}

Contact med control for
other possible treatments:
- PRBC transfusion
- Ascorbic acid
- Hyperbaric oxygen
therapy

Supportive care
as needed

Continue
supportive
care

Figure 2: Methemoglobin Levels & Symptoms

<3%: Normal Level

3-12%: Asymptomatic

- Mild cyanosis may be visible

13-20%: Usually asymptomatic unless preexisting
condition present (anemia, heart or lung disease)

20-50%: Mild-Moderate symptoms of hypoxemia

- Fatigue, tachypnea, tachycardia
- Mental status changes
- Nausea, vomiting

50-70%: Severe, life-threatening symptoms

- Respiratory depression
- Shock
- Seizures
- Coma

> 70%: Typically fatal

Figure 3: Key Signs of Methemoglobinemia

- **Refractory hypoxemia:** SpO2 80-90% despite oxygen administration
- **Saturation-PaO2 gap¹:** SpO2 in 80s with normal (or elevated) PaO2
- **Cyanosis-saturation gap:** cyanosis with SpO2 in 80s (generally should not cause cyanosis until SpO2 is much lower ~ 60%)
- **Chocolate-Brown blood:** blood that remains dark when exposed to air (normal deoxygenated blood will absorb oxygen and turn red)

Figure 4: Drugs/Supplements that can cause Serotonin Syndrome

- SSRIs:** citalopram (Celexa), fluoxetine (Prozac), escitalopram (Lexapro), paroxetine (Paxil), sertraline (Zoloft)
- SNRIs:** duloxetine (Cymbalta), venlafaxine (Effexor XR)
- Bupropion** (Zyban, Wellbutrin SR, Wellbutrin XL)
- TCAs:** amitriptyline, nortriptyline (Pamelor)
- MAOIs:** isocarboxazid (Marplan), phenelzine (Nardil)
- Anti-migraine:** carbamazepine (Tegretol), valproic acid and triptans
- Analgesia:** codeine, fentanyl, hydrocodone, oxycodone, tramadol
- Illicit drugs:** LSD, ecstasy, cocaine, amphetamines
- Herbal supplements:** St. John's wort, ginseng, nutmeg
- Cough/cold meds** containing dextromethorphan
- Anti-nausea:** metoclopramide (Reglan), ondansetron (Zofran)
- Misc:** Linezolid (Zyvox), Lithium, Ritonavir (Norvir)

NOTES:

1. Do not utilize iSTAT for ABG analysis. iSTAT measures the pO2 and then calculates the predicted oxygen saturation and thus will report a falsely elevated oxygen saturation.
2. Glucose 6 phosphate dehydrogenase (G6PD) deficiency is a hereditary condition in which red blood cell hemolysis occurs when the body is exposed to certain foods, drugs, infections or stress. G6PD deficiency symptoms are more common in males (particularly African Americans and those from certain parts of Africa, Asia, and the Mediterranean).
3. Methylene Blue in G6PD deficiency will cause hemolysis
4. Methylene Blue can cause Serotonin Syndrome in pt's taking serotonergic medications
5. If pt's condition improves with Methylene Blue administration and then symptoms return (likely due to a long-acting agent [i.e. Dapsone]), repeat doses of MB may be required. Pt may also benefit from a MB continuous infusion (0.1-0.25 mg/kg/hr) **PER MED CONTROL**
6. Large total doses (~ 7mg/kg) of Methylene Blue can cause Methemoglobinemia